

MASS ADAS L3 — Concept of Operations (ConOps) v0.1

2026-05-21

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Concept of Operations (ConOps) — MASS L3 Tactical Decision Layer — STUB

Field	Value
Document ID	MASS-L3-TDL-CONOPS-001
Version	v0.1-stub
Date	2026-05-19
Status	STUB — placeholder for D1.8 deliverable
Architecture Baseline	v1.1.2
Development Plan	v3.0
CCS i-Ship Requirement	CG-136 (ConOps is mandatory for AIP)

Purpose

This Concept of Operations describes the MASS ADAS L3 Tactical Decision Layer from the perspective of its users (ROC, Captain, Shore Operator), the operational environment, and the system’s intended behaviour across its operational envelope. It is a mandatory artefact for CCS i-Ship (Nx, Ri/Ai) AIP submission per CG-136.

Required Sections (to be filled)

- ☐ **§1 Scope & System Overview**
 - ☐ 1.1 System boundaries (L3 in MASS stack)
 - ☐ 1.2 Key operational capabilities
 - ☐ 1.3 Relationship to other MASS layers (L1/L2/L4/L5)
- ☐ **§2 User Roles & Responsibilities**
 - ☐ 2.1 Remote Operator / Commander (ROC) — MASS Code level 3–4 operation
 - ☐ 2.2 Captain (onboard, override authority)
 - ☐ 2.3 Shore Operator (monitoring, limited intervention)
 - ☐ 2.4 Role transitions under MASS Code mode changes
- ☐ **§3 Operational Scenarios**
 - ☐ 3.1 Normal open-water transit
 - ☐ 3.2 COLREGs encounter resolution (head-on, crossing, overtaking)
 - ☐ 3.3 Multi-vessel encounters (3+ vessels)
 - ☐ 3.4 Degraded sensor conditions (fog, radar outage)

- ☐ 3.5 Out-of-ODD situation (return to safe state)
- ☐ 3.6 Emergency takeover ($TMR \leq 60s$)
- ☐ **§4 Operational Envelope**
 - ☐ 4.1 ODD parameter ranges (visibility, traffic, waterway)
 - ☐ 4.2 Permitted vessel types (FCB baseline → multi-vessel)
 - ☐ 4.3 Environmental limits (wind/wave/current)
 - ☐ 4.4 Transition to DEGRADED / CRITICAL modes
- ☐ **§5 User Interface Concept**
 - ☐ 5.1 SAT-1/2/3 transparency framework
 - ☐ 5.2 Decision information panels (CPA/TCPA/Rule)
 - ☐ 5.3 Alarm and alert philosophy
 - ☐ 5.4 Takeover request (TOR) modalities
- ☐ **§6 Safety Philosophy**
 - ☐ 6.1 Doer-Checker architecture
 - ☐ 6.2 Safe state definition and entry conditions
 - ☐ 6.3 Graceful degradation path
- ☐ **§7 Assumptions & Constraints**
- ☐ **§8 Reference Documents**

Reference Documents

- Architecture Design v1.1.2: docs/Design/Architecture
Design/MASS_ADAS_L3_TDL_架构设计报告.md
- V&V Plan: docs/Design/V&V_Plan/00-vv-strategy-v0.1.md
- HMI Design (M8): docs/Design/Detailed Design/M8/01-detailed-design.md
- SIL Frontend: docs/Design/SIL/v1.0-unified/03-sil-frontend-design.md
- HAZID RUN-001: docs/Design/HAZID/RUN-001-kickoff.md
- CCS CG-136 (ConOps)
- IMO MASS Code (MSC 110/111)

Development Timeline

Version	Date	Scope	Status
v0.1 stub	2026-05-19	Framework + placeholder sections	 STUB
v0.2 draft	2026-07-31 (D2.8)	Full draft for Phase 2 review	 Planned
v1.0 final	2026-08-31 (D3.8)	AIP-ready version	 Planned